

that minterms 1 and 9 are contained in the prime-implicant $xy'z$.

The completed prime-implicant table is inspected for columns containing only a single cross. In this example, there are four minterms whose columns have a single cross: 1, 4, 8 and 10. Minterm 1 is covered by prime-implicant $xy'z$, i.e. the selection of prime-implicant $xy'z$ guarantees that minterm 1 is included in the function. Similarly, minterm 4 is covered by prime-implicant $w'xz'$, and minterms 8 and 10, by prime-implicant wz .

Prime-implicants that cover minterms with a single cross in their column are called essential prime-implicants. To enable the final simplified expression to contain all the minterms, we have no alternative but to include essential prime-implicants. A check mark is placed in the table next to the essential prime-implicants to indicate that they have been selected.

Next we check each column whose minterm is covered by the selected essential prime-implicants. For example, the selected prime-implicant $xy'z$ covers minterms 1 and 9. A check is inserted in the bottom of the columns. Similarly, prime-implicant $w'xz'$ covers minterms 4 and 6, and wz covers minterms 8, 9, 10 and 11. $\odot$

Inspection of the prime-implicant table shows that the selection of the essential prime-implicants covers all the minterms of the function 7 and 15. These two minterms must be included by the selection of one or more prime-implicants. In this example, it is clear that prime-implicant xyz covers both minterms and is therefore the one to be selected. We have thus found the minimum set of prime-implicants whose sum gives the required minimized function:

$$F = xy'z + w'xz' + wz + xyz$$